

# CLASS 10 MATH

Advanced Board & Olympiad Prep

*Quadratics, Trigonometry, Statistics & 100 Mastery Problems*

# Chapter 1: Real Numbers & Polynomials

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## SECTION A: CORE CONCEPTS

1. State Euclid's Division Lemma.

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2. Find the HCF of 867 and 255 using Euclid's division algorithm.

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3. Explain why  $3 \times 5 \times 7 + 7$  is a composite number.

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4. Find the zeroes of the quadratic polynomial  $3x^2 - x - 4$  and verify the relationship between zeroes and coefficients.

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5. If  $\alpha$  and  $\beta$  are zeroes of  $p(x) = 2x^2 + 5x + 1$ , find the value of  $\alpha + \beta + \alpha\beta$ .

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## SECTION B: REAL WORLD APPLICATIONS

1. An army contingent of 616 members is to march behind an army band of 32 members in a parade. The two groups are to march in the same number of columns. What is the maximum number of columns in which they can march?

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2. There is a circular path around a sports field. Sonia takes 18 minutes to drive one round of the field, while Ravi takes 12 minutes for the same. Suppose they both start at the same point and at the same time, and go in the same direction. After how many minutes will they meet again at the starting point?

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3. A manufacturer produces two types of electronics. If the production is modeled by  $P(x) = x^2 - 10x + 21$ , find the production milestones (zeroes of the polynomial).

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## Chapter 2: Algebra (Linear & Quadratic Equations)

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### SECTION A: CORE CONCEPTS

1. Solve the pair of linear equations by substitution:  $x + y = 14$ ,  $x - y = 4$ .

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2. For what value of  $k$  will the following pair of linear equations have no solution?  $3x + y = 1$ ;  $(2k-1)x + (k-1)y = 2k+1$ .

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3. Find the roots of the quadratic equation  $2x^2 - x + 1/8 = 0$ .

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4. Find the discriminant of the equation  $3x^2 - 2x + \frac{1}{3} = 0$  and hence find the nature of its roots.

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5. Find two consecutive odd positive integers, sum of whose squares is 290.

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## SECTION B: REAL WORLD APPLICATIONS

1. A train travels 360 km at a uniform speed. If the speed had been 5 km/h more, it would have taken 1 hour less for the same journey. Find the speed of the train.

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2. Five years ago, Nuri was thrice as old as Sonu. Ten years later, Nuri will be twice as old as Sonu. How old are Nuri and Sonu?

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3. The sum of a two-digit number and the number obtained by reversing the digits is 66. If the digits of the number differ by 2, find the number.

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# Chapter 3: Trigonometry & Triangles

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## SECTION A: CORE CONCEPTS

1. Given  $\cot A = 8$ , find  $\sin A$  and  $\sec A$ .

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2. Evaluate:  $\frac{2 \tan 30^\circ}{1 + \tan^2 30^\circ}$ .

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3. Prove the identity:  $(\sec A - \cos A)(\cot A + \tan A) = \tan A \sec A$ .

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4. In  $\triangle ABC$ , right-angled at  $B$ ,  $AB = 24$  cm,  $BC = 7$  cm. Determine  $\sin A$ ,  $\cos A$ .

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5. Prove that:  $\frac{\sin \theta - 2\sin^3 \theta}{2\cos^3 \theta - \cos \theta} = \tan \theta$ .

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## SECTION B: REAL WORLD APPLICATIONS

1. A tree breaks due to a storm and the broken part bends so that the top of the tree touches the ground making an angle  $30^\circ$  with it. The distance between the foot of the tree to the point where the top touches the ground is 8 m. Find the height of the tree.

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2. From a point on the ground, the angles of elevation of the bottom and the top of a transmission tower fixed at the top of a 20 m high building are  $45^\circ$  and  $60^\circ$  respectively. Find the height of the tower.

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3. As observed from the top of a 75 m high lighthouse from the sea-level, the angles of depression of two ships are  $30^\circ$  and  $45^\circ$ . If one ship is exactly behind the other on the same side of the lighthouse, find the distance between the two ships.

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# Chapter 4: Coordinate Geometry, Mensuration & Stats

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## SECTION A: CORE CONCEPTS

1. Find the distance between the points  $A(-5, 7)$  and  $B(-1, 3)$ .

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2. Find the coordinates of the point which divides the line segment joining the points  $(4, -3)$  and  $(8, 5)$  in the ratio  $3:1$  internally.

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3. Find the area of a triangle whose vertices are  $(1, -1)$ ,  $(-4, 6)$  and  $(-3, -5)$ .

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4. Find the mean of the following data using the assumed mean method:

Class (10-25, 25-40, 40-55, 55-70, 70-85, 85-100), Frequencies (2, 3, 7, 6, 6, 6).

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5. A bag contains 3 red balls and 5 black balls. A ball is drawn at random from the bag. What is the probability that the ball drawn is red?

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## SECTION B: REAL WORLD APPLICATIONS

1. A metallic sphere of radius 4.2 cm is melted and recast into the shape of a cylinder of radius 6 cm. Find the height of the cylinder.

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2. A tent is in the shape of a cylinder surmounted by a conical top. If the height and diameter of the cylindrical part are 2.1 m and 4 m respectively, and the slant height of the top is 2.8 m, find the area of the canvas used for making the tent.

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3. The following distribution gives the daily income of 50 workers of a factory. Convert the distribution to a 'less than type' cumulative frequency distribution, and draw its ogive.

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# Multi-Concept Combined Practice

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## LEVEL 1: CORE PROFICIENCY (AVERAGE STUDENT) - SET 1

1. Find the HCF and LCM of 12, 15, and 21 using the prime factorization method.

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2. Find the zeroes of the quadratic polynomial  $x^2 - 2x - 8$ .

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3. Solve the pair of linear equations:  $x + y = 14$  and  $x - y = 4$ .

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4. Find the discriminant of the quadratic equation  $2x^2 - 4x + 3 = 0$ .

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5. Find the 10th term of the AP: 2, 7, 12, ...

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6. Evaluate:  $\sin 60^\circ \cos 30^\circ + \sin 30^\circ \cos 60^\circ$ .

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7. Find the distance between the points  $(2, 3)$  and  $(4, 1)$ .

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8. A tangent  $PQ$  at a point  $P$  of a circle of radius 5 cm meets a line through the centre  $O$  at a point  $Q$  so that  $OQ = 12$  cm. Length  $PQ$  is?

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9. Find the volume of a right circular cone with radius 6 cm and height 7 cm.

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10. A die is thrown once. Find the probability of getting a prime number.

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# Multi-Concept Combined Practice

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## LEVEL 1: CORE PROFICIENCY (AVERAGE STUDENT) - SET 2

1. Prove that  $\sqrt{2}$  is an irrational number.

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2. Find a quadratic polynomial whose zeroes are  $-3$  and  $2$ .

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3. Check whether the pair of equations  $5x - 4y + 8 = 0$  and  $7x + 6y - 9 = 0$  is consistent or inconsistent.

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4. Find the roots of the quadratic equation  $x^2 - 3x - 10 = 0$  by factorization.

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5. Find the sum of the first 22 terms of the AP: 8, 3, -2, ...

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6. Given  $15 \cot A = 8$ , find  $\sin A$  and  $\sec A$ .

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7. Find the coordinates of the point which divides the join of  $(-1, 7)$  and  $(4, -3)$  in the ratio  $2:3$ .

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8. If tangents  $PA$  and  $PB$  from a point  $P$  to a circle with centre  $O$  are inclined to each other at angle of  $80^\circ$ , find  $\angle POA$ .

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9. A metallic sphere of radius 4.2 cm is melted and recast into the shape of a cylinder of radius 6 cm. Find the height of the cylinder.

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10. One card is drawn from a well-shuffled deck of 52 cards. Find the probability of getting a king of red colour.

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# Multi-Concept Combined Practice

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## LEVEL 2: ADVANCED APPLICATION (TOP 10%) - SET 1

1. Show that any positive odd integer is of the form  $4q + 1$  or  $4q + 3$ , where  $q$  is some integer.

2. If  $\alpha$  and  $\beta$  are the zeroes of the polynomial  $f(x) = x^2 - 5x + k$  such that  $\alpha - \beta = 1$ , find the value of  $k$ .

3. Solve the equations:  $\frac{2}{x} + \frac{3}{y} = 13$  and  $\frac{5}{x} - \frac{4}{y} = -2$ .

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4. A motor boat whose speed is  $18$  km/h in still water takes  $1$  hour more to go  $24$  km upstream than to return downstream to the same spot. Find the speed of the stream.

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5. Which term of the AP:  $3, 15, 27, 39, \dots$  will be  $132$  more than its  $54$ th term?

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6. Prove that:  $(\sin A + \csc A)^2 + (\cos A + \sec A)^2 = 7 + \tan^2 A + \cot^2 A$ .

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7. Find the area of the triangle formed by the points  $(2, 3)$ ,  $(-1, 0)$ , and  $(2, -4)$ .

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8. Prove that the lengths of tangents drawn from an external point to a circle are equal.

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9. A toy is in the form of a cone of radius 3.5 cm mounted on a hemisphere of same radius. The total height of the toy is 15.5 cm. Find the total surface area of the toy.

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10. Find the mean of the following distribution: Class intervals 0-10, 10-20, 20-30, 30-40, 40-50 with frequencies 5, 8, 12, 10, 5.

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# Multi-Concept Combined Practice

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## LEVEL 2: ADVANCED APPLICATION (TOP 10%) - SET 2

1. Use Euclid's division lemma to show that the cube of any positive integer is of the form  $9m$ ,  $9m+1$  or  $9m+8$ .

2. Divide  $3x^2 - x^3 - 3x + 5$  by  $x - 1 - x^2$ , and verify the division algorithm.

3. A fraction becomes  $\frac{1}{3}$  when 1 is subtracted from the numerator and it becomes  $\frac{1}{4}$  when 8 is added to its denominator. Find the fraction.

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4. Find two consecutive odd positive integers, sum of whose squares is 290.

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5. Find the sum of all two-digit odd positive numbers.

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6. The shadow of a tower standing on a level ground is found to be 40 m longer when the Sun's altitude is  $30^\circ$  than when it is  $60^\circ$ . Find the height of the tower.

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7. Find the ratio in which the y-axis divides the line segment joining the points  $(5, -6)$  and  $(-1, -4)$ .

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8. Prove that the parallelogram circumscribing a circle is a rhombus.

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9. A solid consisting of a right circular cone of height 120 cm and radius 60 cm standing on a hemisphere of radius 60 cm is placed upright in a right circular cylinder full of water. Find the volume of water left.

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10. The median of the following data is 525. Find the values of  $x$  and  $y$ , if the total frequency is 100.

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# Multi-Concept Combined Practice

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## LEVEL 3: LOGICAL REASONING (TOP 1%) - SET 1

1. If  $\alpha$  and  $\beta$  are the zeroes of the quadratic polynomial  $f(x) = ax^2 + bx + c$ , then evaluate  $\frac{\alpha}{\beta} + \frac{\beta}{\alpha}$ .

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2. Solve for  $x$ :  $\frac{1}{a+b+x} = \frac{1}{a} + \frac{1}{b} + \frac{1}{x}$ , where  $a+b \neq 0$ .

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3. Determine the arithmetic progression whose third term is 16 and the 7th term exceeds the 5th term by 12.

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4. Prove that:  $\frac{\tan \theta}{1 - \cot \theta} + \frac{\cot \theta}{1 - \tan \theta} = 1 + \sec \theta \csc \theta$ .

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5. A straight highway leads to the foot of a tower. A man standing at the top of the tower observes a car at an angle of depression of  $30^\circ$ . Six seconds later, the angle is  $60^\circ$ . Find the time taken to reach the foot.

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6. Prove that the ratio of the areas of two similar triangles is equal to the square of the ratio of their corresponding medians.

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7. Find the coordinates of the circumcentre of the triangle whose vertices are  $(8, 6)$ ,  $(8, -2)$  and  $(2, -2)$ .

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8. Two tangents  $TP$  and  $TQ$  are drawn to a circle with centre  $O$  from an external point  $T$ . Prove that  $\angle PTQ = 2\angle OPQ$ .

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9. Water is flowing at the rate of 15 km/h through a pipe of diameter 14 cm into a cuboidal pond which is 50 m long and 44 m wide. In what time will the level rise by 21 cm?

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10. Two dice are thrown simultaneously. What is the probability that the sum of the two numbers appearing on the top is a prime number?

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# Multi-Concept Combined Practice

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## LEVEL 3: LOGICAL REASONING (TOP 1%) - SET 2

1. Show that  $n^2 - 1$  is divisible by 8, if  $n$  is an odd positive integer.

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2. If a polynomial  $x^4 - 6x^3 + 16x^2 - 25x + 10$  is divided by another polynomial  $x^2 - 2x + k$ , the remainder comes out to be  $x + a$ . Find  $k$  and  $a$ .

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3. For what values of  $a$  and  $b$  does the following pair of linear equations have an infinite number of solutions:  $2x + 3y = 7$  and  $(a-b)x + (a+b)y = 3a+b-2$ .

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4. A train travels 360 km at a uniform speed. If the speed had been 5 km/h more, it would have taken 1 hour less for the same journey. Find the speed of the train.

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5. The sum of the third and the seventh terms of an AP is 6 and their product is 8. Find the sum of first sixteen terms of the AP.

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6. If  $\sin \theta + \cos \theta = \sqrt{2}$ , prove that  $\tan \theta + \cot \theta = 2$ .

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7. Prove that in a right triangle, the square of the hypotenuse is equal to the sum of the squares of the other two sides (Pythagoras Theorem).

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8. Find the area of the shaded region if  $ABCD$  is a square of side 14 cm and  $APD$  and  $BPC$  are semicircles.

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9. A bucket is in the form of a frustum of a cone of height 30 cm with radii of its lower and upper ends as 10 cm and 20 cm. Find the capacity of the bucket.

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10. If the mean of the following frequency distribution is 65.6, find the missing frequencies  $f_1$  and  $f_2$ .

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# Multi-Concept Combined Practice

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## LEVEL 4: CRITICAL THINKING (TOP 0.1%) - SET 1

1. Prove that  $\sqrt{p} + \sqrt{q}$  is irrational, where  $p$  and  $q$  are distinct primes.

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2. Solve the system of non-linear equations:  $2^x + 3^y = 17$  and  $2^{x+2} - 3^{y+1} = 5$ .

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3. Find the roots of the equation  $x^2 + x - (a+1)(a+2) = 0$ .

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4. If the  $m^{\text{th}}$  term of an AP is  $\frac{1}{n}$  and the  $n^{\text{th}}$  term is  $\frac{1}{m}$ , show that the sum of  $mn$  terms is  $\frac{1}{2}(mn+1)$ .

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5. Prove the trigonometric identity:  $\frac{\cos A - \sin A + 1}{\cos A + \sin A - 1} = \csc A + \cot A$ .

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6. The angle of elevation of a cloud from a point  $h$  metres above a lake is  $\alpha$  and the angle of depression of its reflection in the lake is  $\beta$ . Prove that the height of the cloud is  $h \frac{\tan \beta + \tan \alpha}{\tan \beta - \tan \alpha}$ .

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7. Let  $ABC$  be an equilateral triangle inscribed in a circle of radius  $R$ . Prove that the sum of the squares of the distances from any point  $P$  on the circle to the vertices is  $6R^2$ .

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8. A circle touches the sides of a quadrilateral  $ABCD$  at  $P, Q, R, S$ . Show that the angles subtended at the centre by a pair of opposite sides are supplementary.

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9. A right triangle whose sides are 3 cm and 4 cm (other than hypotenuse) is made to revolve about its hypotenuse. Find the volume and surface area of the double cone so formed.

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10. What is the probability that a randomly selected leap year has 53 Sundays and 53 Mondays?

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# Multi-Concept Combined Practice

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## LEVEL 4: CRITICAL THINKING (TOP 0.1%) - SET 2

1. Find the condition that the zeroes of the polynomial  $f(x) = x^3 - px^2 + qx - r$  may be in arithmetic progression.

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2. Solve for  $x$ :  $3^{x^2 - 4x + 3} = 1$ .

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3. The ratio of the sum of  $m$  and  $n$  terms of an AP is  $m^2 : n^2$ . Show that the ratio of the  $m^{\text{th}}$  and  $n^{\text{th}}$  term is  $(2m-1) : (2n-1)$ .

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4. If  $\sec \theta + \tan \theta = p$ , prove that  $\sin \theta = \frac{p^2 - 1}{p^2 + 1}$ .

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5. A round balloon of radius  $r$  subtends an angle  $\alpha$  at the eye of the observer while the angle of elevation of its centre is  $\beta$ . Prove that the height of the centre of the balloon is  $r \sin \beta \csc \frac{\alpha}{2}$ .

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6. Prove Ceva's Theorem for cyclic quadrilaterals: The product of the diagonals is equal to the sum of the products of the opposite sides.

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7. Determine the locus of a point which moves such that the sum of the squares of its distances from the axes is equal to 4.

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8. If the radii of the circular ends of a conical bucket which is 45 cm high are 28 cm and 7 cm, find the capacity of the bucket.

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9. A bag contains 5 red balls and some blue balls. If the probability of drawing a blue ball is double that of a red ball, determine the number of blue balls in the bag.

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10. In a frequency distribution, the mean and median are 21 and 22 respectively. Find its mode using the empirical relationship.

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# Multi-Concept Combined Practice

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## LEVEL 5: NATIONAL OLYMPIAD (TOP 10) - SET 1

1. Find the highest power of 10 that divides  $1000!$ .

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2. Let  $x_1, x_2, x_3$  be the roots of the equation  $x^3 - 6x^2 + 11x - 6 = 0$ . Find the value of  $x_1^3 + x_2^3 + x_3^3$ .

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3. Solve in integers the equation  $x^2 - y^2 = 2026$ .

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4. An infinite geometric progression has sum 2 and the sum of the squares of its terms is  $\frac{4}{3}$ . Find the common ratio.

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5. In triangle  $ABC$ , the medians  $AD$  and  $BE$  are perpendicular to each other. If  $AC = 19$  and  $BC = 22$ , find the length of  $AB$ .

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6. Find the minimum value of  $f(x) = |x - 1| + |x - 2| + |x - 3| + |x - 4|$ .

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7. A point  $P$  is inside a square  $ABCD$  such that  $PA = 1$ ,  $PB = 2$ ,  $PC = 3$ . Find the measure of  $\angle APB$ .

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8. Evaluate the infinite product:  $3^{1/3} \times 9^{1/9} \times 27^{1/27} \times \dots$

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9. A sphere of radius  $R$  contains 4 identical spheres of radius  $r$  which are mutually tangent and also touch the outer sphere. Find  $r$  in terms of  $R$ .

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10. Find the number of ways to arrange the letters of the word 'COMBINATORICS' such that no two vowels are adjacent.

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# Multi-Concept Combined Practice

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## LEVEL 5: NATIONAL OLYMPIAD (TOP 10) - SET 2

1. Determine all prime numbers  $p$  such that  $p^2 + 2$  is also a prime.

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2. Find the remainder when  $x^{2026} + x^{2025} + \dots + x + 1$  is divided by  $x^2 - 1$ .

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3. Solve the system of equations:  $x + y + z = 1$ ,  $x^2 + y^2 + z^2 = 2$ ,  $x^3 + y^3 + z^3 = 3$ . Find  $x^4 + y^4 + z^4$ .

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4. Let  $S_n$  denote the sum of the first  $n$  terms of an AP. If  $S_{3n} = 3 S_{2n}$ , show that  $S_{4n} = 4 S_{3n}$ .

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5. Prove that in any triangle  $ABC$ ,  $\cos A + \cos B + \cos C \leq \frac{3}{2}$ .

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6. A circle of radius 1 rolls without slipping around the outside of a regular hexagon of side 2. Find the area of the region bounded by the path of the centre of the circle.

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7. If  $x, y, z$  are positive real numbers such that  $xyz = 1$ , prove that  $(x+y)(y+z)(z+x) \geq 8$ .

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8. A regular icosahedron (20 faces) has numbers 1 to 20 written on its faces. You can add 1 to the numbers on any 5 faces that share a common vertex. Can you make all numbers equal?

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9. Find the exact value of  $\sin 18^\circ$ .

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10. How many polynomials  $P(x)$  of degree 5 satisfy  $P(k) = 1/(k+1)$  for  $k = 0, 1, 2, 3, 4, 5$ ?

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